The 2nd Meeting of the Special Interest Group in Non-Equilibrium Molecular Dynamics

Edward Smith, Matthew Borg and James Ewen

06/09/21

Online

Plan for Today

Plan on the website: https://fluids.ac.uk/sig/NonEqmMD

Name	Title/Subject	Start (BST)	Chair
	Welcome	08:00	
Ed Smith	**Introduction**	08:20	
Peter Daivis	Energy and temperature in NEMD simulations	08:30	
Billy Todd	Pumping of nano-confined water with rotating electric fields	09:00	
Jun Zhang	Molecular Dynamics and Multiscale simulation of Droplet Wetting and Evaporation	09:30	
	Discussion	10:00	
Karl Travis	New Dolls tensor-based algorithm for studying liquid fragmentation and the tensile test for solids.	10:30	Matthew Borg
Patrick Ilg	Surface rheology of polymers at the liquid-vapour interface	10:45	
Suman Chakraborty	Evaporation of Saline Aqueous Nano-Droplets: The Coffee-Ring Effect and Beyond	11:00	
Paola Carbone	Thermodynamic properties of the graphene/electrolyte interface	11:30	
	Lunch	12:00	

Plan for Today

Plan on the website: https://fluids.ac.uk/sig/NonEqmMD

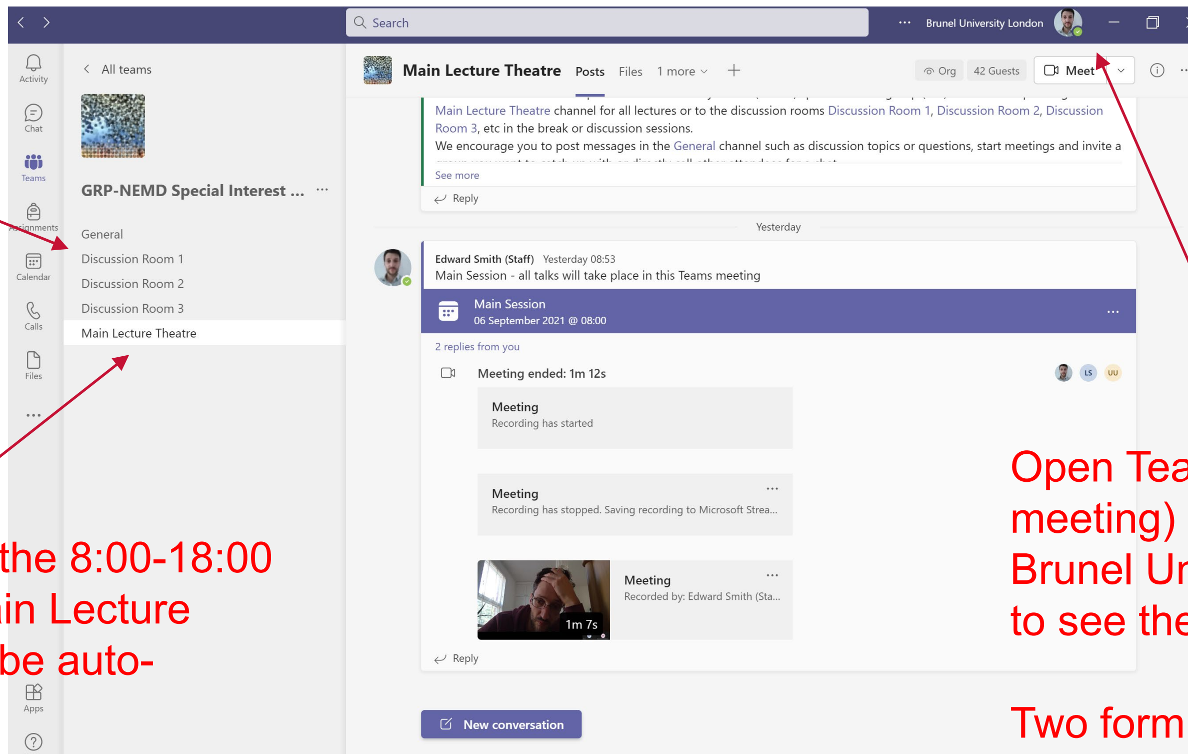
	Lunch	12:00	
Philip Camp	Using non-equilibrium molecular-dynamics simulations to study problems in industry	13:00	
Alessio Lavino	Surface topography effects on pool boiling via non-equilibrium molecular dynamics simulations	13:30	James Ewen
Ozgur Yazaydin	Concentration Gradient Driven Molecular Dynamics to Model Gas and Liquid Phase Separations in Membranes	13:45	
Jesper Hansen	On nanoscale polarization	14:00	
	Discussion	14:30	
Kaihang Shi	Can we define a unique microscopic pressure in inhomogeneous fluids?	15:00	
Jagjeevan Bhamra	Interfacial Bonding Controls Friction in Diamond-Rock Contacts	15:30	
Dimitrios Mathas	Evaluation of methods for viscosity simulations of lubricants at different temperatures and pressures: a case study on PAO-2	15:45	
James Sprittles	Noisy Interfacial Nanoflows	16:00	
Lorenzo Botto	Graphene nanohydrodynamics: non-equilibrium MD simulation of graphene nanoplatelets suspended in sheared liquids	16:30	Ed Smith
Sergey Karabasov	A thermostat-consistent fully coupled molecular dynamics – generalised fluctuating hydrodynamics model for non-equilibrium flows	17:00	
	Discussion	17:15	
Snook Prize	An overview of the 2021 $1000 Snook prize offered by Bill and Carol Hoover	17:40	
	Close	18:00	

Using the GRP-NEMD SIG Team

If you joined by the meeting link, you may not be in “the Team”

Note a “Team” has “channels” where “meetings” occur

Parallel discussion can occur in various channels (like conference rooms)

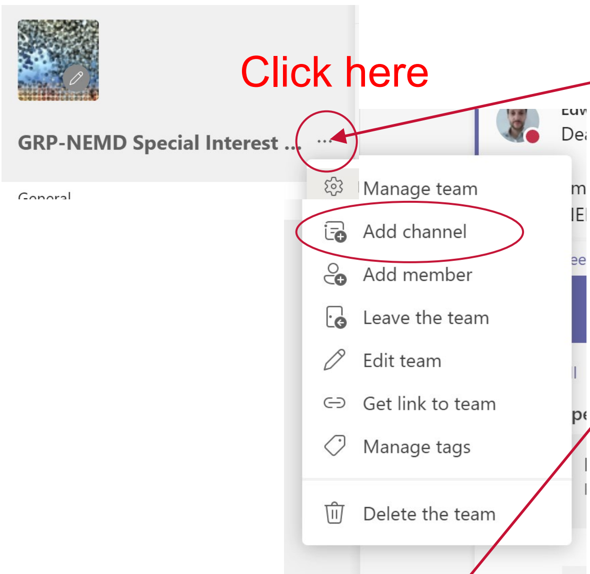


All talks will be in the 8:00-18:00 meeting in the Main Lecture Theatre. This will be auto-recorded

Open Teams (not in a meeting) and change to Brunel University (guest) to see the Team.

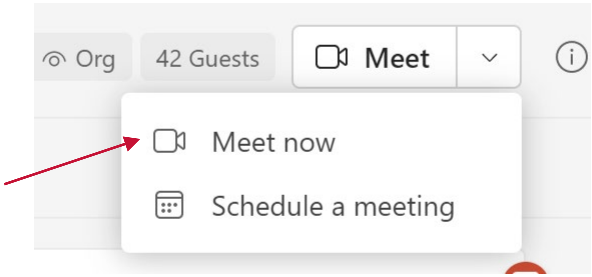
Two form authentication may be needed (App on phone of text/call)

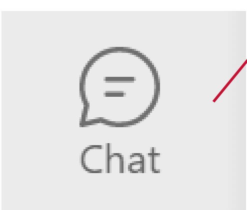
Making the Most of Teams

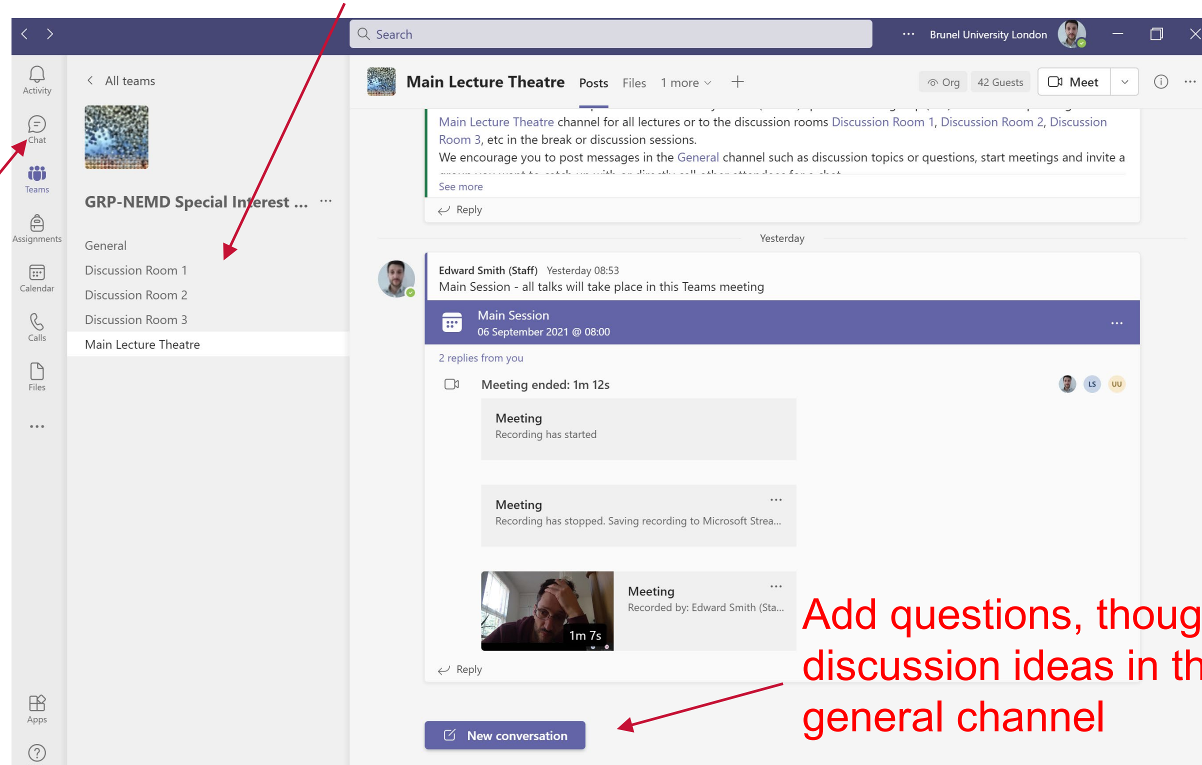


Click here

Create new meetings in discussion rooms and ask fellow attendees.
Add new channels for new discussions







Chat or call fellow attendees for one on one discussions

Add questions, thoughts, discussion ideas in the general channel

Direct Links

- If you cannot get Teams to work, use the direct link to the meeting in the main lecture theatre (for all talks) :

https://teams.microsoft.com/l/meetup-join/19%3a224d643941f24409a72a4979477e015f%40thread.tacv2/1630569208147?context=%7b%22Tid%22%3a%224cad97b1-5935-4103-a866-57ad98a1517e%22%2c%22Oid%22%3a%220ffd7534-f205-4108-9efa-d14887f5f851%22%7d

- You will only be able to join any discussions in the main lecture theatre meeting and not see other channels

What is the UK Fluids Network (UKFN)?

UK FLUIDS NETWORK

- EPSRC Funded (EP/N032861/1) in 2016 (ended 29th Feb 2020)
  - Aims to keep the UK leading in fluid mechanics
  - To engage as a group with industry
  - To build leadership within the community
  - To facilitating inter-disciplinary research

- Unlike other networks, aims to target the "many emerging areas, which have the biggest potential to create major step changes, fall between the cracks."

- Support 40+ Special Interest Groups (SIGs) that address industrial, scientific, and societal challenges outside existing joint efforts.

www.fluids.ac.uk

What is a Special Interest Group (SIG)?

https://fluids.ac.uk/sig

SIG SRV Talks Resources Directory Gallery Connect Edward Smith

Special Interest Groups

SIG meeting calendar | Join a SIG | SIG Booklet

All SIGs

There are 41 Special Interest Groups spanning 63 institutions.

Acoustofluidics

Aeroacoustics

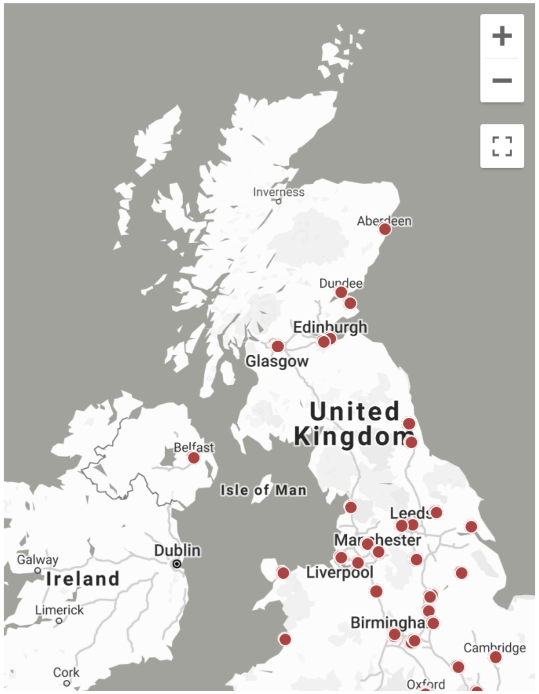
Biologically active fluids

Boundary layers and complex rotating flows

Challenges in cardiovascular flow modelling

Combustion science, technology and applications

Computational Astrophysical Fluid Dynamics (CAFD) (NEW)



All SIGs

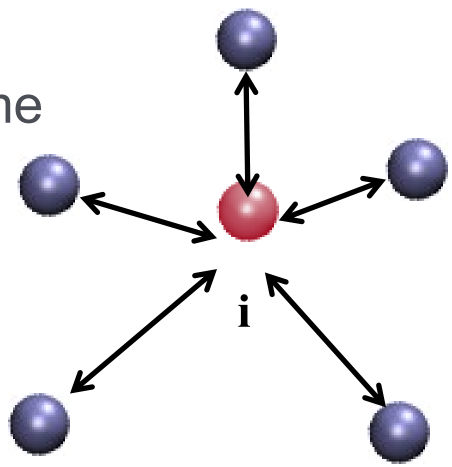
Institutions

University of Bolton
University of Cambridge
University of Durham
University of Edinburgh
University of Glasgow
Heriot-Watt University
University of Leeds
Loughborough University
Northumbria University
Nottingham Trent University
University of Southampton
University of Strathclyde
University of Warwick
University of Bath
University of Bristol
Brunel University London
Imperial College London
Keele University
University of Leicester

What is NEMD - Molecular Dynamics

Discrete molecules in continuous space

- Molecular position evolves continuously in time
- Position and velocity from acceleration



$$\ddot{\boldsymbol{r}}_i \rightarrow \dot{\boldsymbol{r}}_i$$
$$\dot{\boldsymbol{r}}_i \rightarrow \boldsymbol{r}_i(t)$$

Acceleration obtained from forces

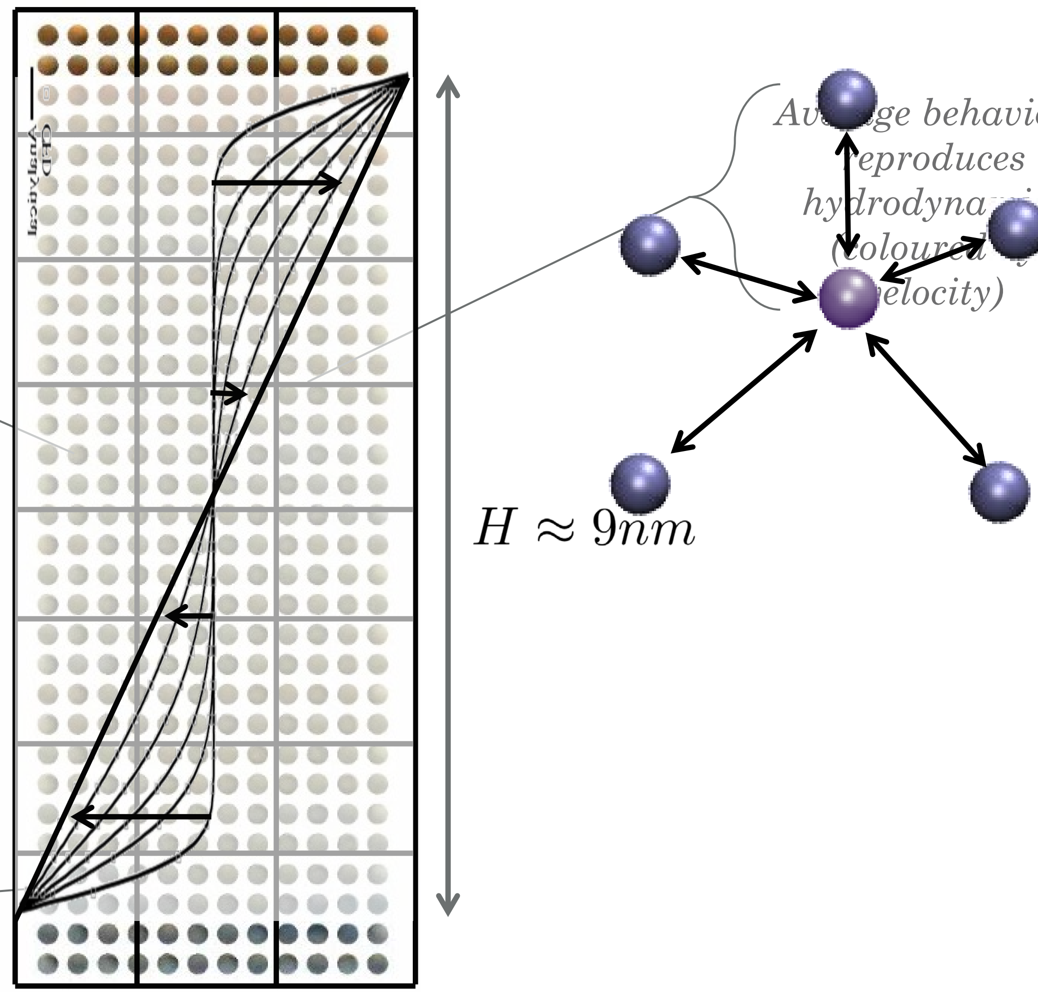
- Governed by Newton’s law for an N-body system
- Point particles with pairwise interactions only

$$m_i \ddot{\boldsymbol{r}}_i = \mathbf{F}_i = \sum_{i \neq j}^{N} \boldsymbol{f}_{ij} = \sum_{i \neq j}^{N} \boldsymbol{\nabla} \Phi_{ij} \qquad \mathit{\Phi}(r_{ij}) = 4\epsilon \left[\left(\frac{\ell}{r_{ij}} \right)^{12} - \left(\frac{\ell}{r_{ij}} \right)^{6} \right]$$

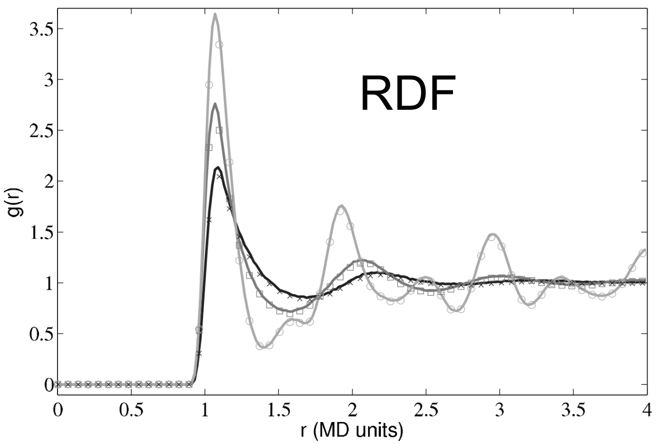
Non-Equilibrium Molecular Dynamics

Liquid structure causes viscosity

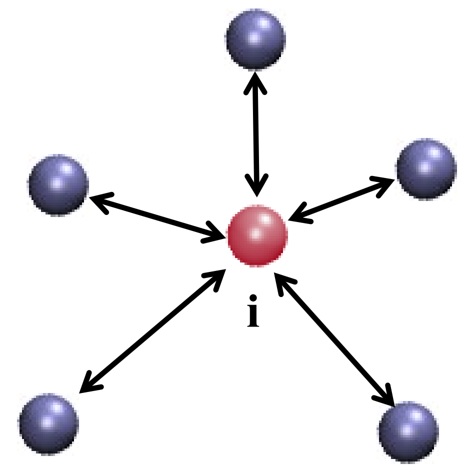
Stick-slip near walls



Non-Equilibrium Molecular Dynamics



Liquid Structure is key



Fluctuation Theorem

$$\frac{P(\dot{S} = A)}{P(\dot{S} = -A)} = e^{At}$$

Stress/Pressure

$$\Pi_{xy} = \underbrace{\sum_{i=1}^{N} \left\langle m_i v_{xi} v_{yi} dS_{yi} \right\rangle}_{\text{Kinetic}} + \underbrace{\frac{1}{2} \sum_{i=1}^{N} \sum_{j \neq i}^{N} \left\langle f_{xij} dS_{yij} \right\rangle}_{\text{Configurational}}$$

Viscosity

$$\mu = \frac{V}{k_B T} \int_0^t \Pi_{xy}(\tau) \Pi_{xy}(0) \; d\tau$$

Liouville's Equation

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \sum_{i=1}^{N} \left[\frac{\partial f}{\partial q_i} \dot{q}_i + \frac{\partial f}{\partial p_i} \dot{p}_i \right]$$

Surface Tension

$$\gamma = \int \left[\Pi_N - \Pi_T \right] dr$$

Strong theoretical developments to understand the flow of molecules

NEMD SIG Subject and Scope

- Non-equilibrium molecular dynamics (NEMD) is the study of fundamental fluid flow using MD
- The non-equilibrium betrays the subject's origins, away from thermodynamic equilibrium due to
  - Wall motion, temperature and pressure gradients
  - Interfaces, e.g. a liquid-vapour coexistence.
- This is **Fluid Dynamics**, traditionally modelled by continuum differential equations
  - Molecular models allow a different way to simulate these problems
  - NEMD has a strong statical mechanics theoretical underpinning
  - Fundamental expertise has the potential to revolutionise nano-scale fluid dynamics – an area of growing importance for industry.

Our SIG in Non-Equilibrium Molecular Dynamics (NEMD)

- Please sign up to the UKFN for future announcements

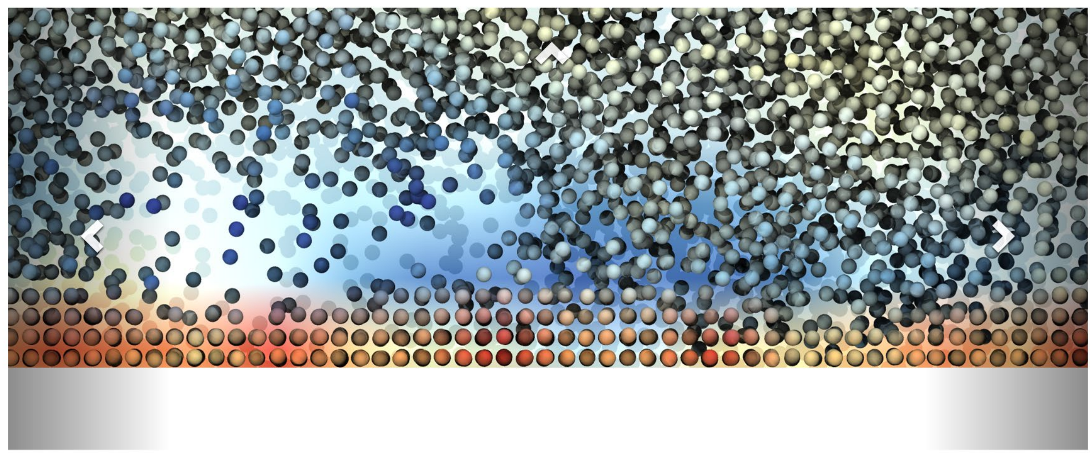
- The mailing list from this meeting could be used for a future international NEMD conference

https://fluids.ac.uk/sig/NonEqmMD

www.fluids.ac.uk/sig/NonEqmMD

SIG SRV Talks Resources Directory Gallery Connect

Edward Smith



Non-equilibrium Molecular Dynamics (NEW)

Description

Non-equilibrium molecular dynamics (NEMD) is the study of fundamental fluid flow using molecular simulation - 'non-equilibrium' because the system is driven away from thermodynamic equilibrium by wall motion, temperature and pressure gradients or

Tweets

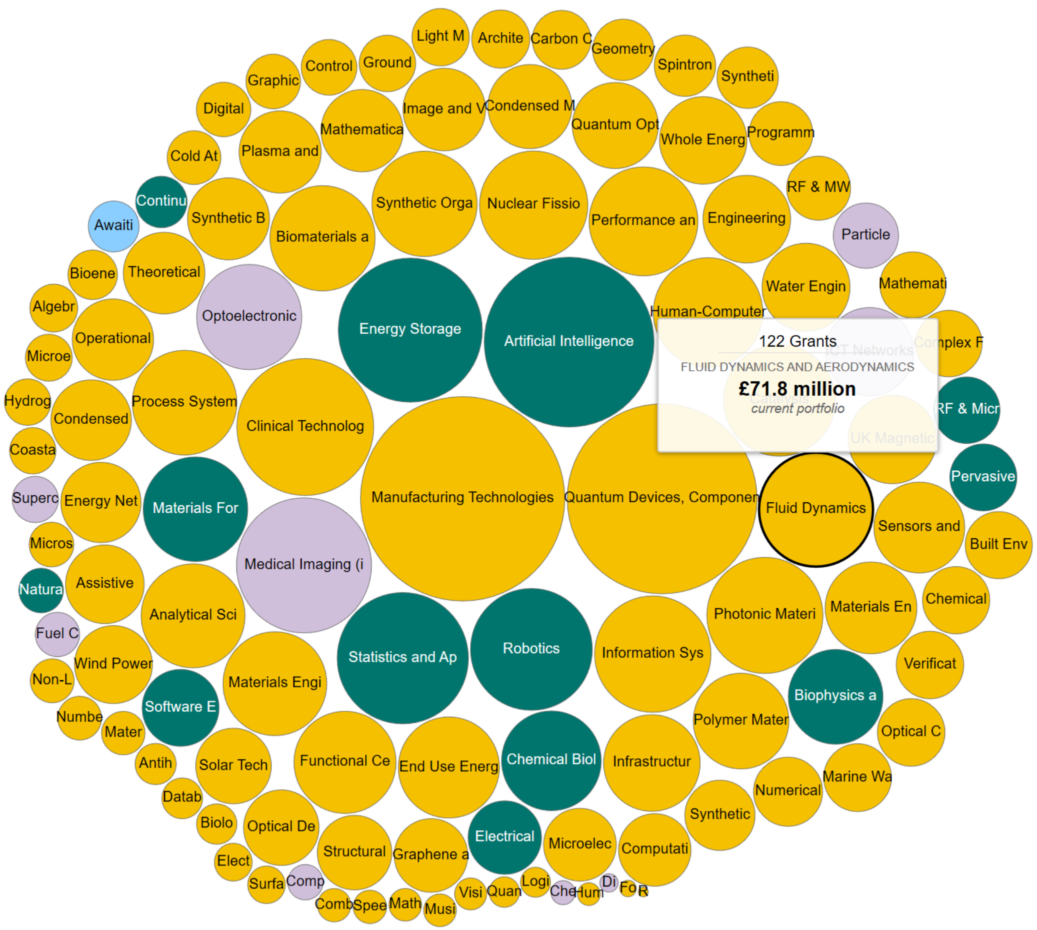
@UKFluidsNetwork

UK Fluids Network Retweeted

SIG Objectives

- Expose the UKFN to benefits of non-equilibrium molecular dynamics theory,
  - Molecular underpinning of viscous/viscoelastic flow, liquid-vapour interface, nucleation of bubbles
  - Origins of instability, turbulence and links to the second law
  - Detailed atomistic understanding of tribological, biological or chemical structures in flows
- In exchange, the NEMD community will gain
  - A new perspective on important fluid flow problems
  - New analysis techniques
  - Links to problems of interest to industry

SIG Objectives



Fluid Dynamics is a major funding area

UK strong in fundamental fluids research & in NEMD – overlap has huge potential

Many important problems need grants and students (a NEMD CDT?)

Discussion at the last meeting

organise follow up meetings ✓

seek extra funding → ?

keep the meetings fairly general

Seek funding from:
- National Nuclear Lab [contact: Mark Bankhead]
- Stat Mech & Thermo Gp of RSC [Chair Fernando Bresme]
- CCP5 (if they still exist!)
- EPSRC. (NETWORK GRANTS)
- Liquids & Complex Fluids Group of IoP. [Chair: Tannie Liverpool]

Consider Joint meeting with other SIGS
- SPH SIG
- Thermal Hydraulics SIG

- E.U. (CECAM)
- Thomas Young Centre

Links to Funding

Molecular Fluid dynamics (MFD)

Steps to CDT – No MD training

→ MD Supervisor pool

→ Extend partners (tribo, BP, P&G nuder)

→ Exciting problems

→ Fill CCP5 vacuum
- Non eq. CANES
- IC (TCM)

- Coating Processes. – moving contact lines.
- Quenching (Polymer) – Phase change
- High-speed electronics cooling – Critical Heat Transfer
- Miniaturization → nanotech → looking at the nano-scale to design "nano-tech"
- Water filtration
- Fracking – Flow in nano/micro pores
- Molecular design

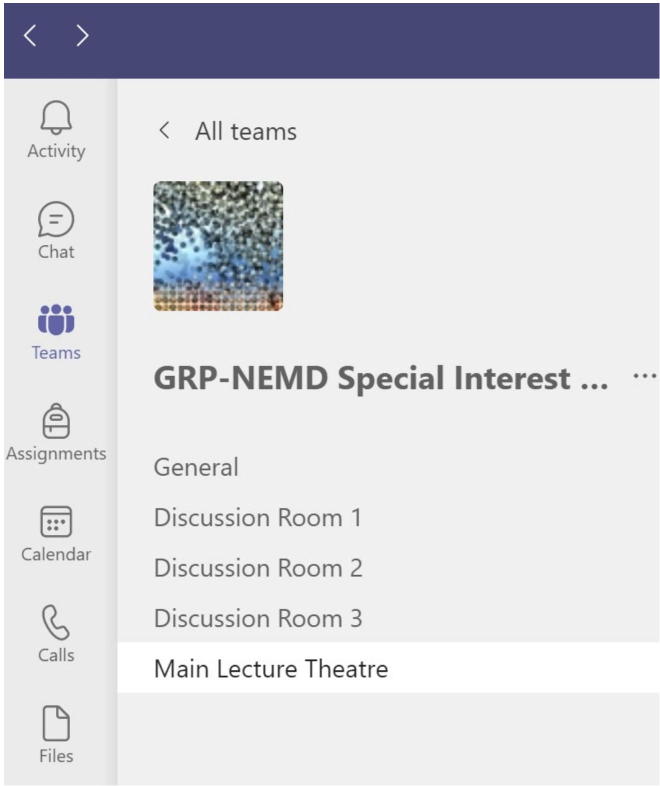
- USER-FRIENDLY SOFTWARE
- REPRODUCIBLE EXAMPLES OF INDUSTRY SOLUTIONS

Send simulation 'delegations' to large companies
- showcase capability
- case studies
- something may spark off, difficult to engineer otherwise

Discussion Today

- In the discussion or lunch sessions, speakers can go to different Discussion Rooms for smaller scale conversations

  Discussions sessions at 10:00-10:30
  Lunch at 12:00-13:00
  Discussion session at 14:30-15:00
  Discussion session at 17:15-17:40



- We also have a chance to discuss topics, please suggest these in the meeting room chat, add to the general channel or email the organisers, e.g.

1. *How much does the atomistic description (up to electronic structure) really affects macroscopic properties such as contact angle, surface tension or friction. Do these effects cancel out at macroscopic level?*
2. *Is it really worth trying to couple the scales? Are there examples showing a multiphysics model is needed?*
3. *How do we establish the more theoretic/blue skies NEMD as a funding area, in the same way fundamental turbulence is studied?*
4. *How do we strengthen links and deliver useful result to industry from MD models?*